Arduino code

```
#include

#include


LiquidCrystal_I2C lcd(0x27, 16, 2);

// Pin Configuration

#define TRIG 2

#define ECHO 3

#define SET_BUTTON 10

#define MODE_BUTTON 11

#define BUZZER 12

#define RELAY 13


// Tank Settings

const float TANK_HEIGHT = 20.0;

const int LOW_LEVEL = 30;

const int HIGH_LEVEL = 85;


// Relay Setting

const bool RELAY_ACTIVE_LOW = false;


// System Variables

bool autoMode = true;
```

```
bool manualPump = false;

bool pumpOn = false;

bool buzzerOn = false;


bool oldSet = HIGH;

bool oldMode = HIGH;


float distance = -1;

int waterLevel = 0;


unsigned long lastPrint = 0;

unsigned long lastLCD = 0;


// Ambulance Siren Variables

unsigned long lastSiren = 0;

bool sirenHigh = false;


// Pump Control

void controlPump(bool state) {


  pumpOn = state;

  if (RELAY_ACTIVE_LOW) {

    digitalWrite(RELAY, state ? LOW : HIGH);

  } else {
```

```
    digitalWrite(RELAY, state ? HIGH : LOW);
  }
}

// Ambulance Buzzer Control
void controlBuzzer(bool state) {

 buzzerOn = state;

 if (!state) {
   noTone(BUZZER);
   sirenHigh = false;
   return;
 }

 if (millis() - lastSiren >= 500) {

   lastSiren = millis();
   sirenHigh = !sirenHigh;

   if (sirenHigh) {
     tone(BUZZER, 5000);
   } else {
     tone(BUZZER, 1200);
   }
```

```
  }
}

// Distance Measurement
float getDistance() {
  digitalWrite(TRIG, LOW);
  delayMicroseconds(3);

  digitalWrite(TRIG, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG, LOW);

  unsigned long duration =
    pulseIn(ECHO, HIGH, 30000);

  if (duration == 0) {
   return -1;
  }

  return duration * 0.0343 / 2.0;
}

// LCD Display
void displayLCD(bool sensorOK) {
```

```
  lcd.setCursor(0, 0);

  if (sensorOK) {
    lcd.print("Water:");
    lcd.print(waterLevel);
    lcd.print("%      ");
  } else {
    lcd.print("Sensor Error    ");
  }

  lcd.setCursor(0, 1);

  lcd.print("Pump:");
  lcd.print(pumpOn ? "ON " : "OFF");

  lcd.print(autoMode ? " AUTO " : " MAN  ");
}

// Setup
void setup() {
  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);

  pinMode(SET_BUTTON, INPUT_PULLUP);
```

```
pinMode(MODE_BUTTON, INPUT_PULLUP);

pinMode(BUZZER, OUTPUT);
pinMode(RELAY, OUTPUT);

controlPump(false);
controlBuzzer(false);

Serial.begin(9600);

lcd.init();
lcd.backlight();

lcd.setCursor(0, 0);
lcd.print("WATER TANK");

lcd.setCursor(0, 1);
lcd.print("MONITOR SYSTEM");

Serial.println("WATER TANK MONITOR STARTED");

delay(1500);
  lcd.clear();
}

// Main Loop
```

```
void loop() {

  // 1. Read Ultrasonic Sensor
  distance = getDistance();
  bool sensorOK =
     (distance >= 2 && distance <= TANK_HEIGHT);

if (sensorOK) {

  waterLevel = round(
    ((TANK_HEIGHT - distance) / TANK_HEIGHT) * 100
  );
  waterLevel = constrain(waterLevel, 0, 100);
}

// 2. MODE Button
bool modeButton = digitalRead(MODE_BUTTON);

if (modeButton == LOW && oldMode == HIGH) {
  autoMode = !autoMode;
  manualPump = false;

delay(200);
```

```
}

oldMode = modeButton;

// 3. SET Button
bool setButton = digitalRead(SET_BUTTON);

if (setButton == LOW && oldSet == HIGH) {

  if (!autoMode && sensorOK) {

    manualPump = !manualPump;

  }

  delay(200);
}

oldSet = setButton;

// 4. Pump and Buzzer Control
if (!sensorOK) {

  manualPump = false;

  controlPump(false);

  controlBuzzer(false);

} else {
```

```
if (autoMode) {

  if (waterLevel <= LOW_LEVEL) {
    controlPump(true);
  }

  if (waterLevel >= HIGH_LEVEL) {
    controlPump(false);
  }

} else {

  controlPump(manualPump);
  }

  // Ambulance Siren at 85%
  controlBuzzer(waterLevel >= HIGH_LEVEL);
}

// 5. LCD Update
if (millis() - lastLCD >= 500) {

  lastLCD = millis();

  displayLCD(sensorOK);
```

```
}

// 6. Serial Monitor
if (millis() - lastPrint >= 1000) {
  lastPrint = millis();

Serial.println("----------------------");

if (sensorOK) {

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  Serial.print("Water Level: ");
  Serial.print(waterLevel);
  Serial.println("%");

} else {

  Serial.println("Sensor Error!");
}

Serial.print("Pump: ");
Serial.println(pumpOn ? "ON" : "OFF");
```

```
    Serial.print("Mode: ");
    Serial.println(autoMode ? "AUTO" : "MANUAL");

    Serial.print("Buzzer: ");
    Serial.println(buzzerOn ? "ON" : "OFF");
  }

  delay(100);
}
```